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Exp No: 9	**Implementation of SUBNETTING in CISCO Packet Tracer Simulator**

AIM:

To design multiple subnets with a suitable number of hosts, assign static IP addresses, connect the subnets through a router in Cisco Packet Tracer, and observe packet transmission with and without the router.

ALGORITHM / PROCEDURE:

1. Take the network address 192.168.1.0/24 and decide the number of subnets needed (3 departments, so 4 subnets are created by borrowing 2 host bits).
2. The new mask is 255.255.255.192 (/26). Each subnet has 64 addresses, so 62 usable hosts.
3. Open Packet Tracer and place 1 router (2911), 3 switches (2960) and 6 PCs. Connect them with copper straight-through cables as in Fig 1.
4. Assign a static IP address, subnet mask and default gateway to every PC under *Desktop > IP Configuration*.
5. Configure the router interfaces: `interface gig0/0`, `ip address 192.168.1.1 255.255.255.192`, `no shutdown`
6. Repeat step 5 for Gig0/1 and Gig0/2 with the gateway address of subnets B and C.
7. From the PC command prompt use `ping` to test connectivity inside a subnet and across subnets.
8. Remove the router (delete the links to the router) and ping across the subnets again to see the difference.

Fig 1: Network topology with three subnets joined by a router

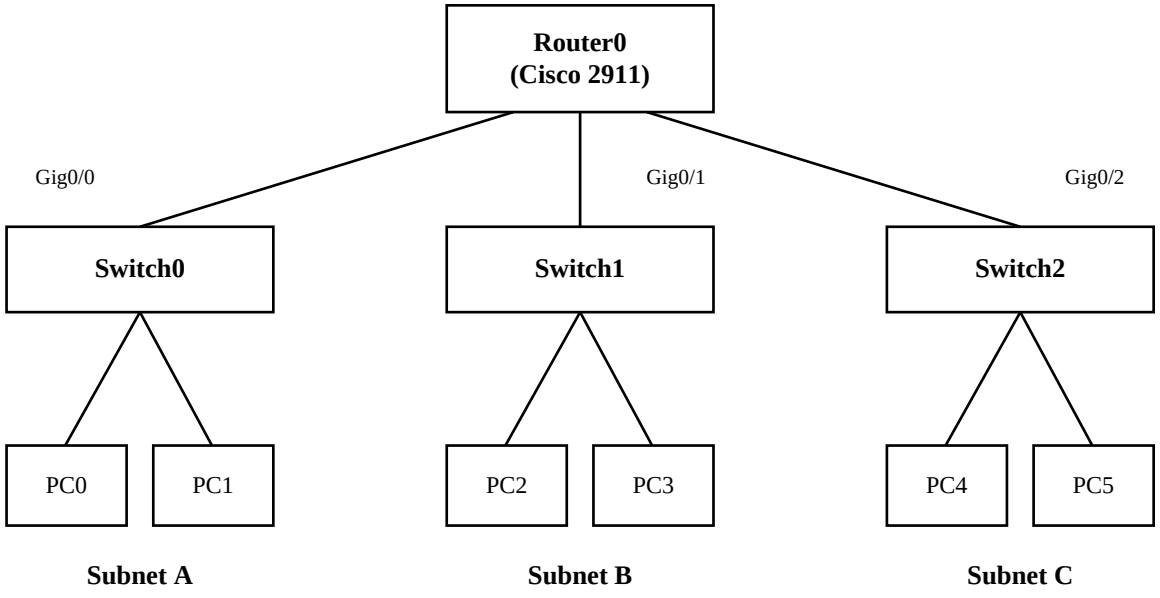
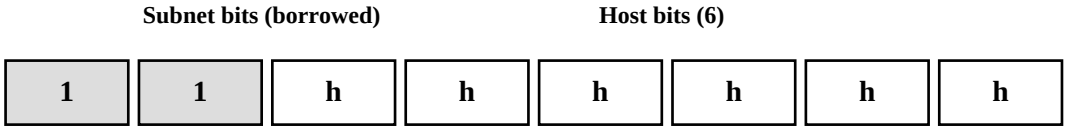


Fig 2: Borrowing 2 bits of the host part for subnetting



192.168.1.0 /24 -> /26 (2 bits borrowed = 4 subnets, 2^6 - 2 = 62 hosts each)

Subnet design:

Subnet	Network ID	Mask	Usable range	Broadcast	Gateway
A (Sales)	192.168.1.0/26	255.255.255.192	.1 - .62	192.168.1.63	192.168.1.1
B (HR)	192.168.1.64/26	255.255.255.192	.65 - .126	192.168.1.127	192.168.1.65
C (IT)	192.168.1.128/26	255.255.255.192	.129 - .190	192.168.1.191	192.168.1.129
Unused	192.168.1.192/26	255.255.255.192	.193 - .254	192.168.1.255	-

IP address table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gig0/0	192.168.1.1	255.255.255.192	-
Router0	Gig0/1	192.168.1.65	255.255.255.192	-
Router0	Gig0/2	192.168.1.129	255.255.255.192	-
PC0	Fa0	192.168.1.2	255.255.255.192	192.168.1.1
PC1	Fa0	192.168.1.3	255.255.255.192	192.168.1.1
PC2	Fa0	192.168.1.66	255.255.255.192	192.168.1.65
PC3	Fa0	192.168.1.67	255.255.255.192	192.168.1.65
PC4	Fa0	192.168.1.130	255.255.255.192	192.168.1.129
PC5	Fa0	192.168.1.131	255.255.255.192	192.168.1.129

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Router configuration:

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.1.129 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#end
Router#write memory
```

OUTPUT:

Router0 CLI - show ip interface brief

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.1.65    YES manual up                    up
GigabitEthernet0/2     192.168.1.129   YES manual up                    up
```

PC0 Command Prompt - same subnet

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

(a) When the subnets are connected via a router

PC0 Command Prompt - ping PC2 (Subnet B) through the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time=3ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router

```
PC>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=2ms TTL=127
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router (contd.)

```
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

(b) When the subnets are not connected (router removed)

PC0 Command Prompt - ping PC2 (Subnet B) without the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

PC0 Command Prompt - ping PC1 (same subnet) without the router

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

OBSERVATION:

Case	Source -> Destination	Result
Router connected	PC0 -> PC1 (same subnet)	Success, TTL = 128
Router connected	PC0 -> PC2 / PC4 (other subnet)	Success, TTL = 127 (one router hop). First packet lost while ARP is resolved.
Router removed	PC0 -> PC1 (same subnet)	Success, switch forwards the frames
Router removed	PC0 -> PC2 / PC4 (other subnet)	Fails, 100% loss. No device can route between subnets.

RESULT:

Thus subnetting of the 192.168.1.0/24 network into /26 subnets was implemented in Cisco Packet Tracer. Hosts in different subnets could communicate only when a router was connected between them, while hosts inside the same subnet could communicate even without the router.